Varunprasaath R. Selvaraju, Ph.D.

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PROFESSIONAL SUMMARY

PhD chemist/materials scientist with 6+ years in polymer and thin-film materials characterization, root-cause failure analysis, and DOE-guided process-property optimization. Experienced translating spectroscopy, thermal analysis, electrochemistry, GPC/MS, and process data into material recommendations, coating/process controls, and technical reports for data-rich engineering teams.

CORE COMPETENCIES

Materials Qualification | Polymer & Coating Characterization | Root Cause Failure Analysis | Thermal Analysis | DOE/SPC | Process Stabilization | Technical Reporting

WORK EXPERIENCE

University of Colorado Boulder
July 2025 - Present

Postdoctoral Researcher

- Lead DOE-guided bio-based polymer process development, defining material-quality readouts, process windows, scale-up specifications, and SOP-ready controls for reproducible execution.
- Translate spectroscopy, thermal-analysis, and process data into technical reports and decision criteria that support material selection, troubleshooting, and process improvement.

University of Colorado Boulder
July 2021 - July 2025

Graduate Researcher

- Mapped degradation, redox instability, and performance failure modes across 15+ functional material variants using FT-IR, UV-Vis-NIR, cyclic voltammetry, TGA/DSC, NMR, GPC, and MS.
- Connected structure, purity, processing history, and stability outcomes to material-design recommendations and first-principles failure hypotheses.

Georgia Institute of Technology
Aug 2019 - July 2021

Graduate Researcher

- Co-led AFOSR-funded thin-film coating process development, linking CVD/surface preparation to morphology, interface quality, optical performance, and reproducible process controls.

ETH Zurich
May 2017 - July 2017

Research Assistant

- Developed and optimized CVD polymer thin-film deposition with DOE strategies and in-situ QCM-D monitoring to quantify growth kinetics and coating reproducibility.

University of Colorado Boulder
2021 - 2025

Laboratory Operations Manager

- Managed characterization/process-development lab operations for a 15-member group; implemented calibration, SOP, safety, and equipment-tracking systems that improved instrument uptime by 30% and strengthened data quality.

Georgia Tech
2019-2021

Teaching Assistant, Organic Chemistry I & Lab

- Mentored 120+ undergraduates; developed supplemental materials improving exam performance by 15%.

KEY PROJECTS

Materials Reliability and Failure-Mode Study
Chemistry of Materials 2025

Mapped degradation, redox instability, and performance failure modes across 15+ functional material variants using orthogonal characterization and stress-test logic.

FT-IR, UV-Vis-NIR, CV, TGA/DSC, NMR, GPC, MS

Bio-Based Polymer Process Qualification
Current

Defined DOE-guided process windows, analytical readouts, reproducibility criteria, and SOP-ready material-quality specifications for polymer scale-up.

DOE, SPC, TGA/DSC, spectroscopy, process documentation

AFOSR Defense Thin-Film Coatings
DoD Funded

Established process-property relationships for CVD/surface-preparation workflows and reproducible high-performance coating conditions.

CVD, QCM-D, surface prep, thin films

EDUCATION

Ph.D. in Chemistry - University of Colorado Boulder 2025

Dissertation: Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials
Advisor: Prof. Seth Marder
(Transferred from Georgia Institute of Technology)

B.S. (Research), Major: Chemistry; Minor: Materials Science - Indian Institute of Science (IISc) 2019

INSPIRE Scholarship for Higher Education (Top 1% nationally)

CERTIFICATIONS

Statistical Process Control (SPC) - Coursera, 2026

TECHNICAL SKILLS

Materials Testing & Characterization: FT-IR, UV-Vis-NIR, ellipsometry, GPC/SEC, LC-MS, GC-MS, NMR, QCM-D, TGA, DSC, cyclic voltammetry; exposure to XRD/SEM-adjacent workflows
Reliability & Failure Analysis: Failure mode identification, root cause analysis, material degradation, stress-test design, anomalous data investigation, design/process mitigation, FMEA principles
Coatings & Materials Platforms: Polymers, biopolymers, organic/organometallic materials, thin films, CVD coatings, surface functionalization, structure-property-processing relationships
Statistical & Process Methods: DOE, SPC, JMP (basic), Python (basic), Excel, process windows, reproducibility studies, scale-up specifications
Technical Execution: SOPs, calibration tracking, technical reports, cross-functional communication, lab safety, equipment uptime, data integrity

JOURNAL PUBLICATIONS

1. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Narrow-Band Electrochromism of a Ferrocene-Substituted Rhodamine B." Chemistry of Materials, 2025, 37, 5558-5568.
2. Mandal, J.; Ramasamy Selvaraju, V.; Yan, W.; Divandari, M.; Spencer, N. D.; Dubner, M. "In Situ Monitoring of SI-ATRP Throughout Multiple Reinitiations Under Flow by Means of a Quartz Crystal Microbalance." RSC Advances, 2018, 8, 20048-20055.
3. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Beyond Rhodamine B: Investigating Alternative Xanthene Cores for Modular Electrochromic Systems." (Manuscript under preparation).

CONFERENCE PRESENTATIONS

Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials, ACS Fall National Meeting, Denver, CO, August 2024.
Electrochemical and Spectroscopic Properties of Substituted Rhodamine Derivatives, Gerischer-Lewerenz Electrochemistry Symposium, Fort Collins, CO, August 2024.
Decoupling Redox and Color for High-Contrast Electrochromics, Excited State Processes Conference, Santa Fe, NM, June 2023.
Electrochromism: Decoupling Color and Redox in Rhodamine-Based Systems, Rocky Mountain Solid State Chemistry Workshop, Boulder, CO, January 2023.

HONORS & AWARDS

Plastic Sustainability Innovation in Material Award | INSPIRE Scholarship (Top 1%, India, 2015) | 97.12 Percentile - IIT JEE Mains (2015) | All India Rank 144 - NEST (2014)